

Section 3: Reproduction and inheritance

- a) Reproduction
- b) Inheritance

a) Reproduction

Students will be assessed on their ability to:

- 3.1 understand the differences between sexual and asexual reproduction
- 3.2 understand that fertilisation involves the fusion of a male and female gamete to produce a zygote that undergoes cell division and develops into an embryo

Flowering plants

- 3.3 describe the structures of an insect-pollinated and a wind-pollinated flower and explain how each is adapted for pollination
- 3.4 understand that the growth of the pollen tube followed by fertilisation leads to seed and fruit formation
- 3.5 understand the conditions needed for seed germination**
- 3.6 understand how germinating seeds utilise food reserves until the seedling can carry out photosynthesis**
- 3.7 understand that plants can reproduce asexually by natural methods (illustrated by runners) and by artificial methods (illustrated by cuttings)

Humans

- 3.8 describe the structure and explain the function of the male and female reproductive systems
- 3.9 understand the roles of oestrogen and progesterone in the menstrual cycle
- 3.10 describe the role of the placenta in the nutrition of the developing embryo**
- 3.11 understand how the developing embryo is protected by amniotic fluid**
- 3.12 understand the roles of oestrogen and testosterone in the development of secondary sexual characteristics.

b) Inheritance

Students will be assessed on their ability to:

- 3.13 understand that the nucleus of a cell contains chromosomes on which genes are located
- 3.14 understand that a gene is a section of a molecule of DNA and that a gene codes for a specific protein
- 3.15 describe a DNA molecule as two strands coiled to form a double helix, the strands being linked by a series of paired bases: adenine (A) with thymine (T), and cytosine (C) with guanine (G)
- 3.16 understand that genes exist in alternative forms called alleles which give rise to differences in inherited characteristics
- 3.17 understand the meaning of the terms: dominant, recessive, homozygous, heterozygous, phenotype, genotype and **codominance**
- 3.18 describe patterns of monohybrid inheritance using a genetic diagram
- 3.19 understand how to interpret family pedigrees
- 3.20 predict probabilities of outcomes from monohybrid crosses
- 3.21 understand that the sex of a person is controlled by one pair of chromosomes, XX in a female and XY in a male
- 3.22 describe the determination of the sex of offspring at fertilisation, using a genetic diagram
- 3.23 understand that division of a diploid cell by mitosis produces two cells which contain identical sets of chromosomes
- 3.24 understand that mitosis occurs during growth, repair, cloning and asexual reproduction
- 3.25 understand that division of a cell by meiosis produces four cells, each with half the number of chromosomes, and that this results in the formation of genetically different haploid gametes
- 3.26 understand that random fertilisation produces genetic variation of offspring
- 3.27 know that in human cells the diploid number of chromosomes is 46 and the haploid number is 23
- 3.28 understand that variation within a species can be genetic, environmental, or a combination of both
- 3.29 understand that mutation is a rare, random change in genetic material that can be inherited
- 3.30 describe the process of evolution by means of natural selection
- 3.31 understand that many mutations are harmful but some are neutral and a few are beneficial
- 3.32 understand that resistance to antibiotics can increase in bacterial populations, and appreciate how such an increase can lead to infections being difficult to control
- 3.33 understand that the incidence of mutations can be increased by exposure to ionising radiation (for example gamma rays, X-rays and ultraviolet rays) and some chemical mutagens (for example chemicals in tobacco).**